

**YEAR 10 SCIENCE EXTENSION
NATURAL AND PROCESSED MATERIALS
ASSIGNMENT 2: CHEMICAL CALCULATIONS**

NAME: SOLUTIONSTotal: 34

1. Calculate the formula mass for the following substances.

NAME	FORMULA	FORMULA MASS (g mol^{-1})
sodium phosphate	Na_3PO_4	163.94
iron (III) nitrate	$\text{Fe}(\text{NO}_3)_3$	241.88
calcium hydrogencarbonate	$\text{Ca}(\text{HCO}_3)_2$	162.12
barium sulphide	BaS	169.36
zinc acetate	$\text{Zn}(\text{CH}_3\text{COO})_2$	183.47

(10)

2. Convert the following to moles or mass as necessary.

(a) 36.2 g of copper (II) nitrate.

(b) 2.34 moles of aluminium hydroxide.

$$\begin{aligned}
 n(\text{Cu}(\text{NO}_3)_2) &= \frac{m}{M} \\
 &= \frac{36.2}{187.57} \quad (1) \\
 &= \underline{0.193 \text{ mol}} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 m(\text{Al}(\text{OH})_3) &= n \times M \\
 &= (2.34)(78.004) \quad (1) \\
 &= \underline{182.5 \text{ g}} \quad (1)
 \end{aligned}$$

(b) 1.34 kg of iron (III) oxide.

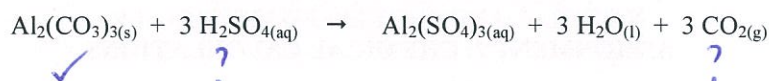
(d) 1.15×10^3 moles of sulphur trioxide.

$$\begin{aligned}
 n(\text{Fe}_2\text{O}_3) &= \frac{m}{M} \\
 &= \frac{1.34 \times 10^3}{159.70} \quad (1) \\
 &= \underline{8.39 \text{ mol}} \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 m(\text{SO}_3) &= n \times M \\
 &= (1.15 \times 10^3)(80.06) \quad (1) \\
 &= \underline{9.21 \times 10^4 \text{ g}} \quad (1)
 \end{aligned}$$

(8)

3. 27.2 g of aluminium carbonate is reacted with excess sulphuric acid. The equation for the reaction is as follows.



- (a) Calculate the mass of sulphuric acid used in the reaction?

$$\begin{aligned} n(\text{Al}_2(\text{CO}_3)_3) &= \frac{m}{M} \\ &= \frac{27.2}{233.99} \quad (1) \\ &= 0.1162 \text{ mol} \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} n(\text{H}_2\text{SO}_4) &= \frac{3}{1} \times n(\text{Al}_2(\text{CO}_3)_3) \quad (1) \\ &= 3(0.1162) \\ &= 0.3486 \text{ mol} \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} m(\text{H}_2\text{SO}_4) &= n \times M \\ &= (0.3486)(98.076) \quad (1) \\ &= \underline{34.19 \text{ g}} \quad (1) \end{aligned}$$

(5)

- (b) What mass of carbon dioxide would be produced?

$$\begin{aligned} n(\text{CO}_2) &= \frac{3}{1} \times n(\text{Al}_2(\text{CO}_3)_3) \quad (1) \\ &= 3(0.1162) \\ &= 0.3486 \text{ mol} \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} m(\text{CO}_2) &= n \times M \\ &= (0.3486)(44.01) \quad (1) \\ &= \underline{15.34 \text{ g}} \quad \left(\frac{1}{2}\right) \end{aligned}$$

(3)

4. Ethanol burns in air according to the following equation.



If 3.85×10^2 g of ethanol burns, calculate:

- (a) the mass of O_2 required.

$$\begin{aligned} n(\text{C}_2\text{H}_5\text{OH}) &= \frac{m}{M} \\ &= \frac{3.85 \times 10^2}{46.068} \quad (1) \\ &= 8.357 \text{ mol} \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} n(\text{O}_2) &= \frac{3}{1} \times n(\text{C}_2\text{H}_5\text{OH}) \quad (1) \\ &= 3(8.357) \\ &= 25.071 \text{ mol} \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} m(\text{O}_2) &= n \times M \\ &= (25.071)(32.00) \quad (1) \\ &= 8.02 \times 10^2 \text{ g} \quad (1) \end{aligned}$$

(5)

- (b) the mass of CO_2 produced.

$$\begin{aligned} n(\text{CO}_2) &= \frac{2}{1} \times n(\text{C}_2\text{H}_5\text{OH}) \quad (1) \\ &= 2(8.357) \\ &= 16.714 \text{ mol} \quad \left(\frac{1}{2}\right) \end{aligned}$$

$$\begin{aligned} m(\text{CO}_2) &= n \times M \\ &= (16.714)(44.01) \quad (1) \\ &= 7.36 \times 10^2 \text{ g} \quad \left(\frac{1}{2}\right) \end{aligned}$$

(3)

